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Reverse image search using centroids of vector embeddings

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U.S. DEPARTMENT OF ENERGY BATTELLE

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What do we want to do?

- Search for images like ones we have (a class)
- Find as many as possible



Problems with searching for a class

- Using just one image is not representative



≠



- Multiple Images = more searches

- Need to represent the class but search faster

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Background

Different methods for Reverse Image Search

- Image analyzing algorithms
- Machine learning ←



Model

Vector Embedding

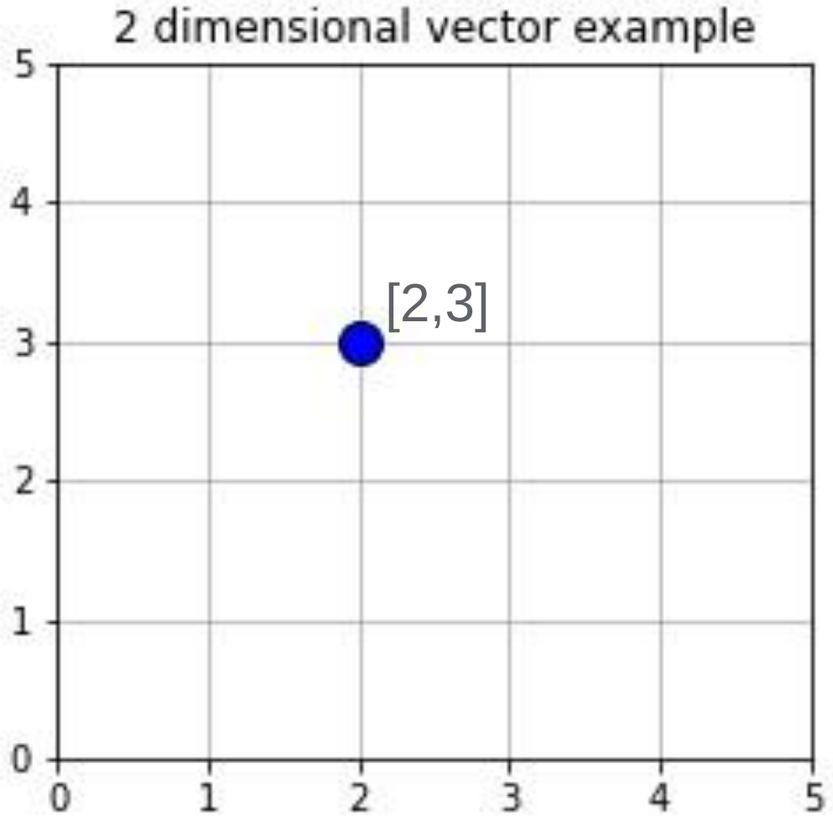
[2,0,7,3,5,9]

Xception: Deep Learning with Depthwise Separable Convolutions, F. Chollet, arXiv 2017, DOI: 10.48550/arXiv.1610.02357, https://doi.org/10.48550/arXiv.1610.02357

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What is a vector embedding?

- Just a list of numbers
- Each number is a dimension
- Produced by a model

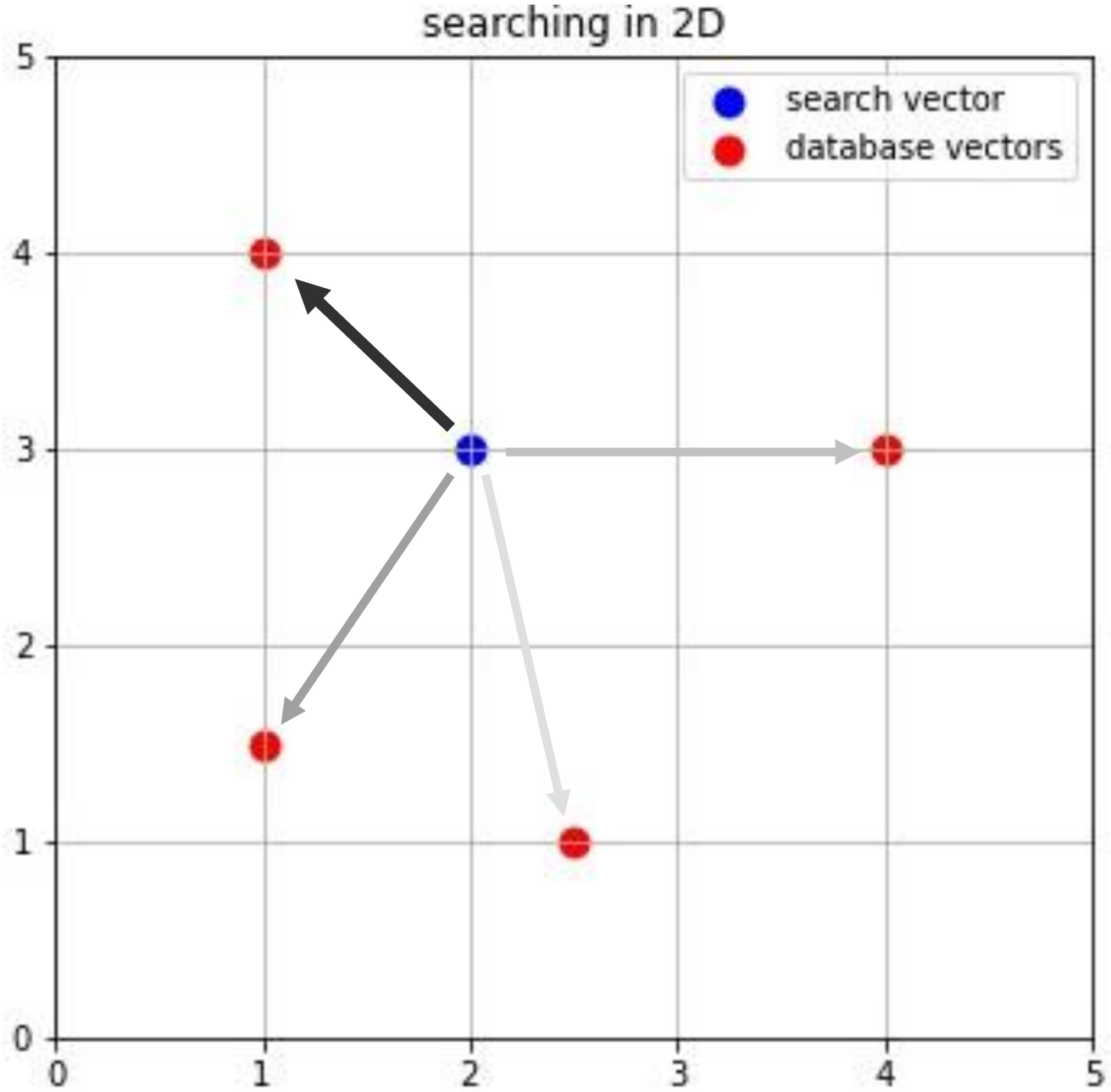


- Our vectors have 2048 dimensions

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Searching

- Vectorize database images
- Vectorize search image
- Compare distances
- Return closest vectors



Revisiting the problem

- We have vectors describing images
- How to represent multiple vectors?
- Try a centroid!

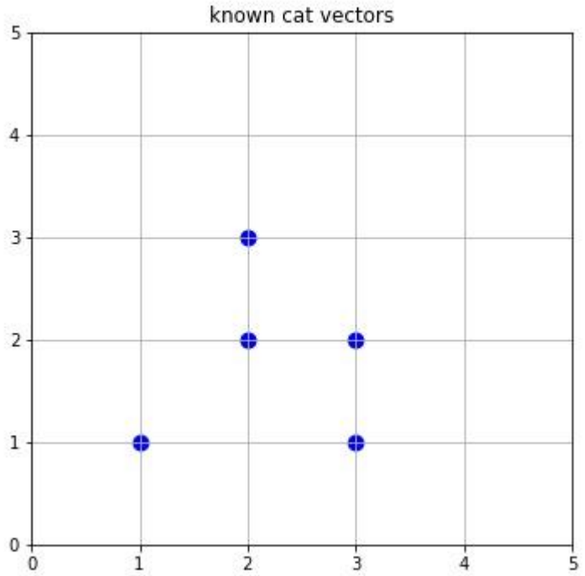


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Model

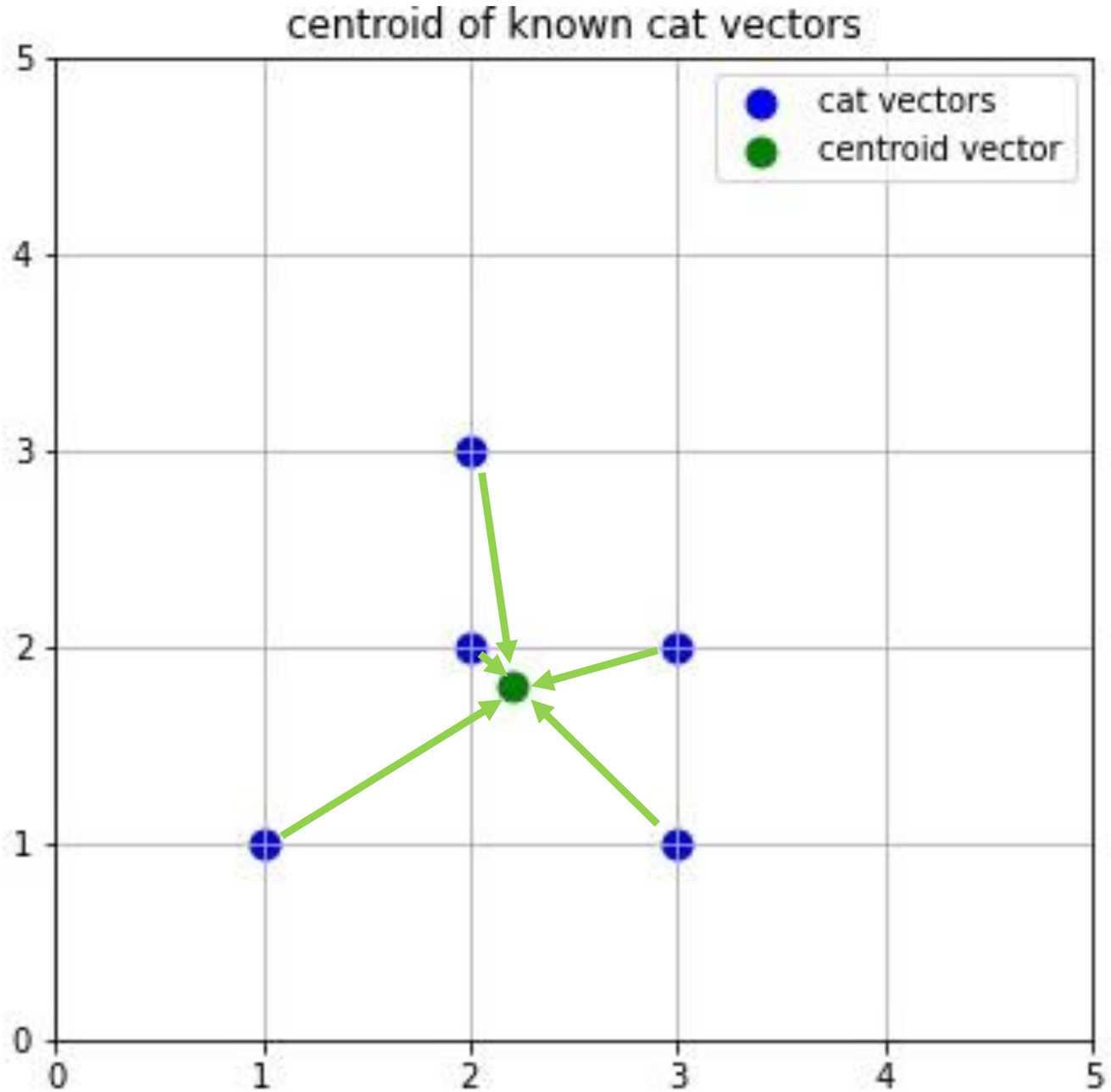
Vector Embeddings



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Centroid

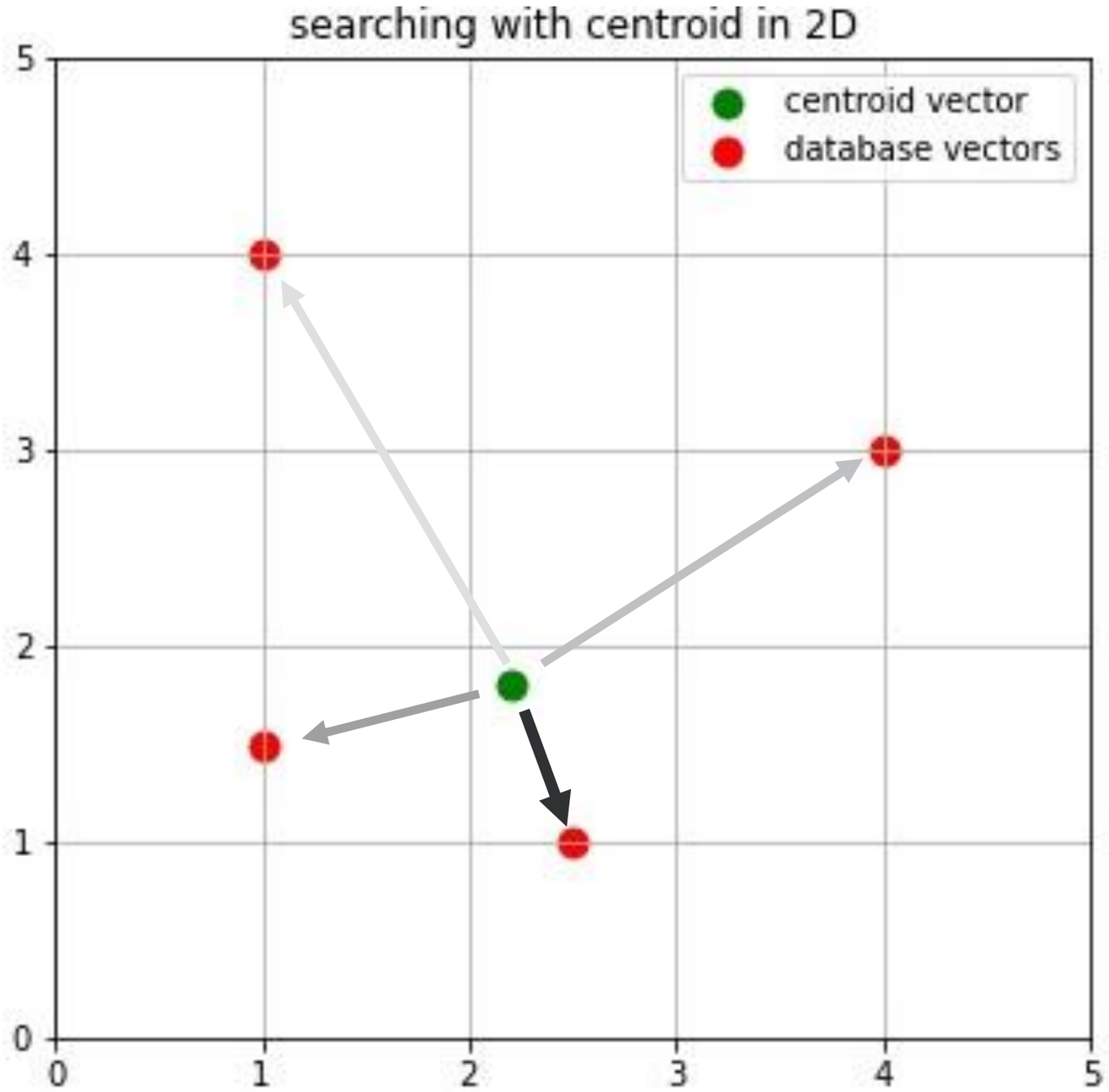
- Take many vectors
- Average their values



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Searching with centroid

- Use centroid from our search vectors
- Compare distances
- Return closest vectors



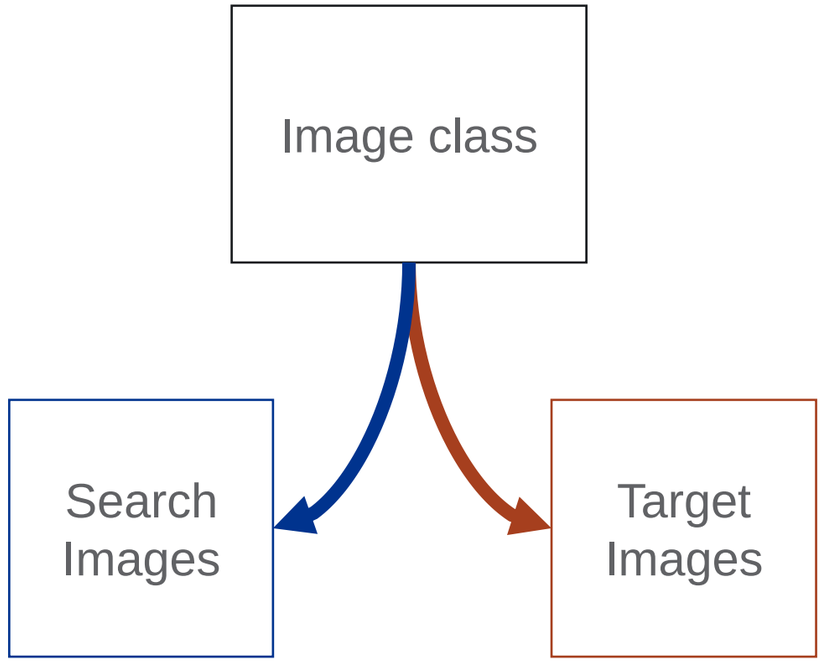
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How to test if a centroid is better?

- Set up a database
- Run the experiment
  - Single image search
  - Centroid
- Measure effectiveness
- Compare the results

Database Setup

- Get dataset of image classes
- For each class, split it
  - Target set -> database
  - Search set -> not in database
- Use search images to find target images



- J. Deng, W. Dong et al., "ImageNet: A large-scale hierarchical image database," 2009 IEEE Conference on Computer Vision and Pattern Recognition, 2009, pp. 248-255, doi: 10.1109/CVPR.2009.5206848.
- **Natural Adversarial Examples,** K. Z. Dan Hendrycks et al., arXiv 2021, DOI: arXiv:1907.07174, https://arxiv.org/abs/1907.07174

Baseline Test Method

For each class

- Search with every search image
  - Measure effectiveness
  - Graph results

Centroid Test Method

For each class

- Search with centroids that use N images
  - Divide search images into groups of N
  - Try different Ns: 3, 5, 10...
  - More images may be better

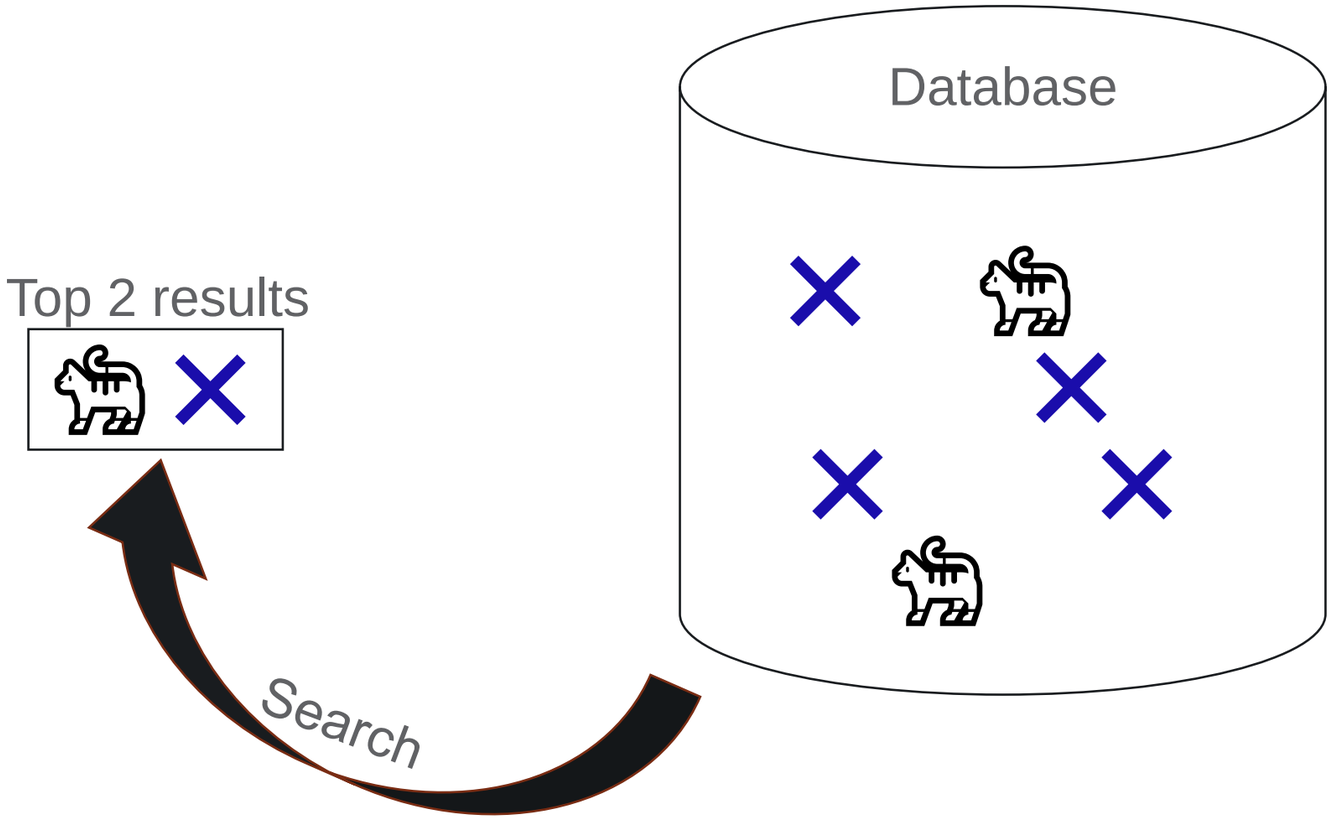
How to measure effectiveness? R-Recall

- R-Recall = % of cats found
- Only check the R closest images
- R = number of database cats
- Usable when target images are known

Evaluation measures (information retrieval), Published by: Wikipedia, Last Updated: 29 November 2022, https://en.wikipedia.org/wiki/Evaluation_measures_(information_retrieval)#Recall

R-Recall Example

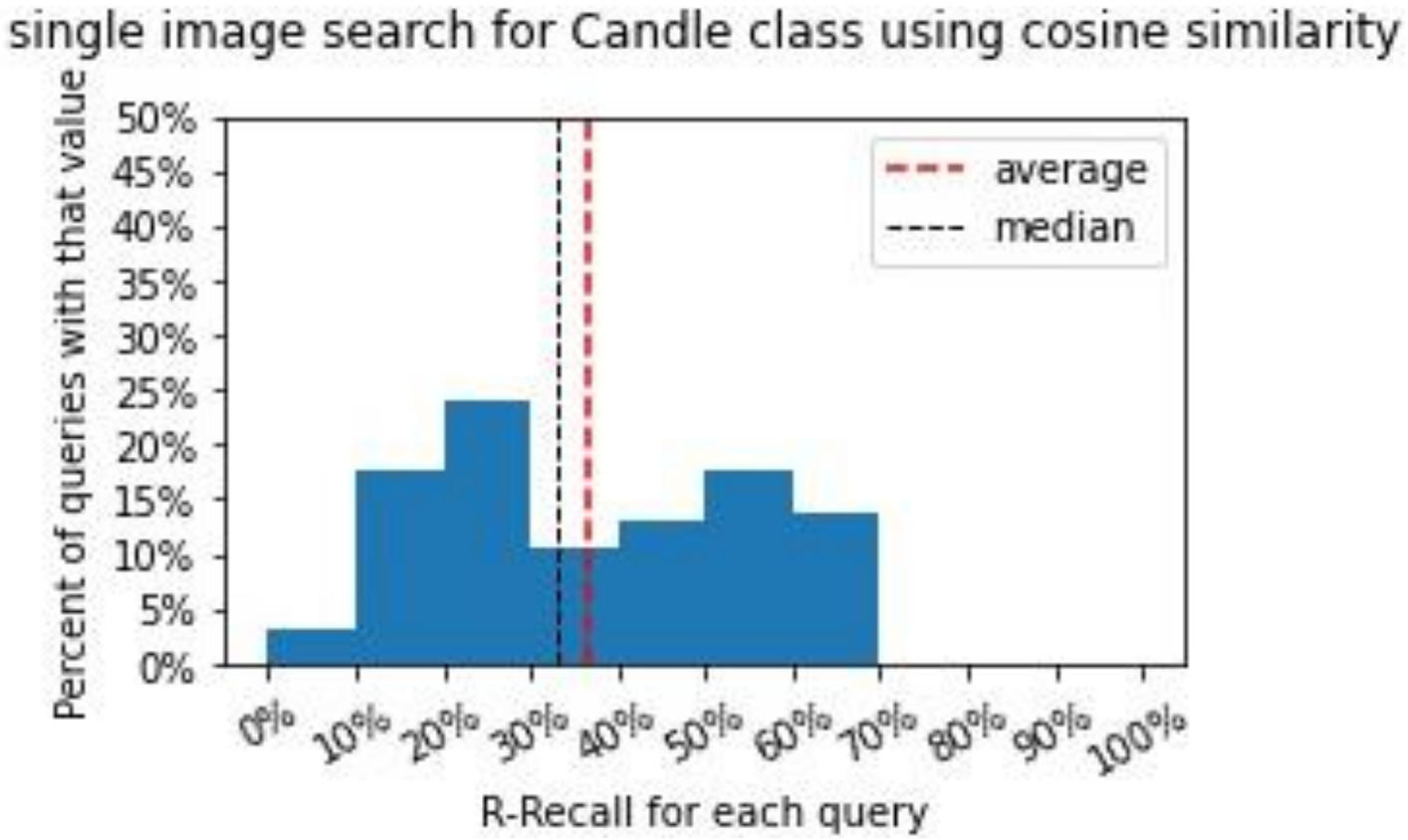
- R-Recall = 1 found ÷ 2 possible = 50%



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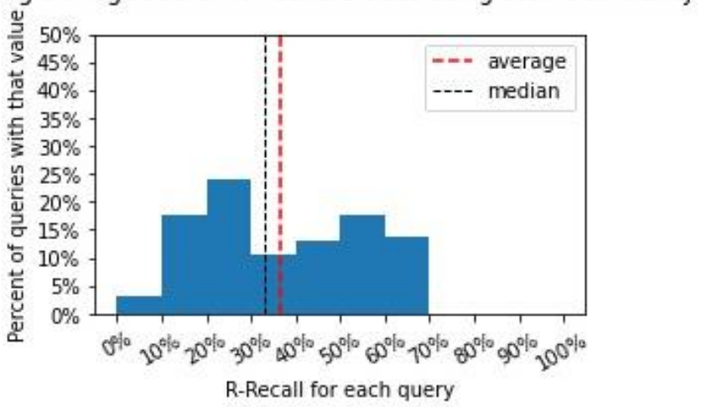
Baseline results, One class

- 124 searches
- X : R-Recall
- Y : Fraction of searches with R Recall value

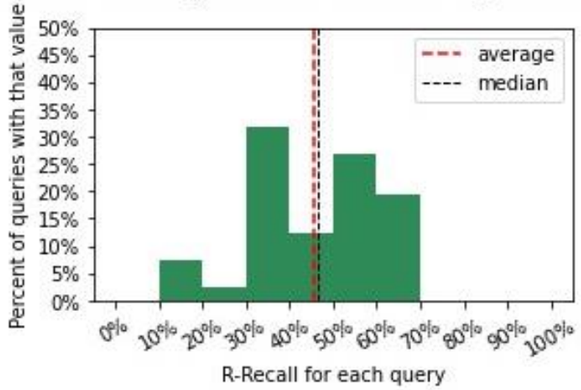


Comparison of Results for One Class (Candle)

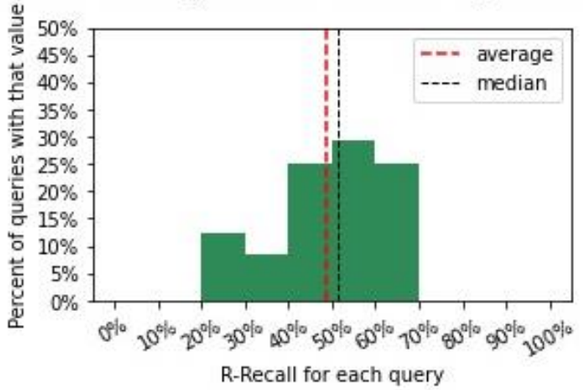
single image search for Candle class using cosine similarity



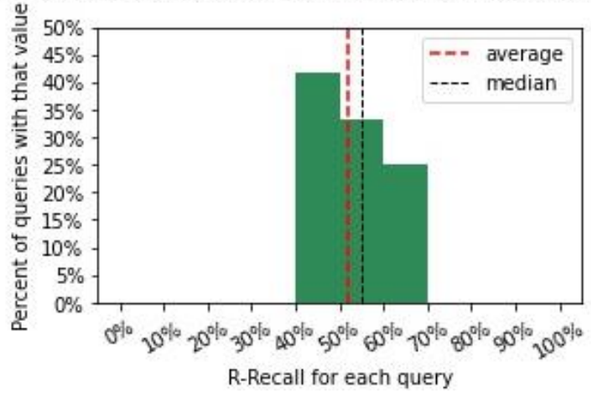
centroids of 3 images for Candle class using cosine similarity



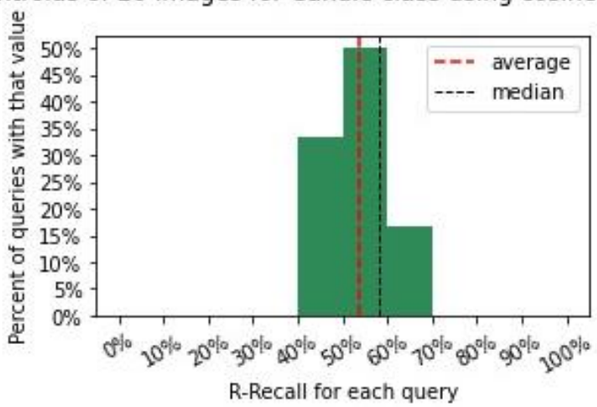
centroids of 5 images for Candle class using cosine similarity



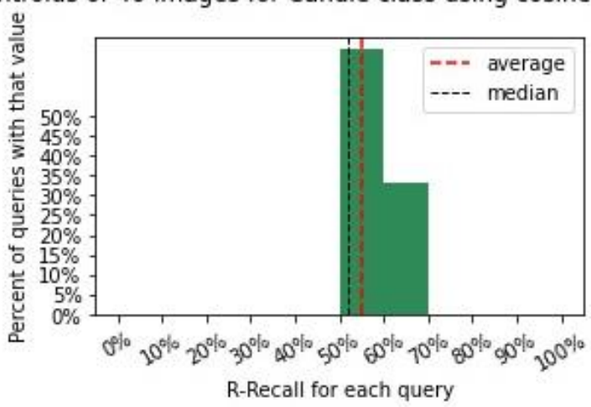
centroids of 10 images for Candle class using cosine similarity



centroids of 20 images for Candle class using cosine similarity

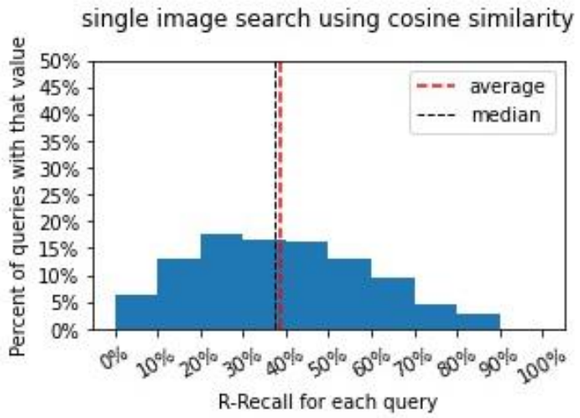


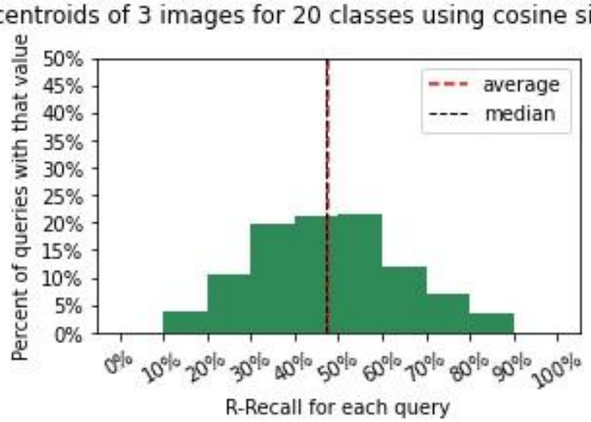
centroids of 40 images for Candle class using cosine similarity

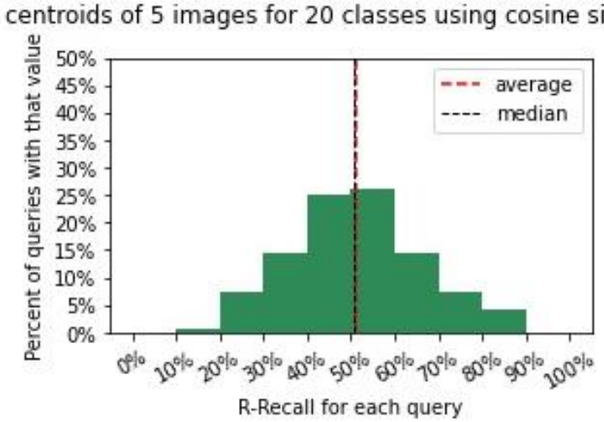


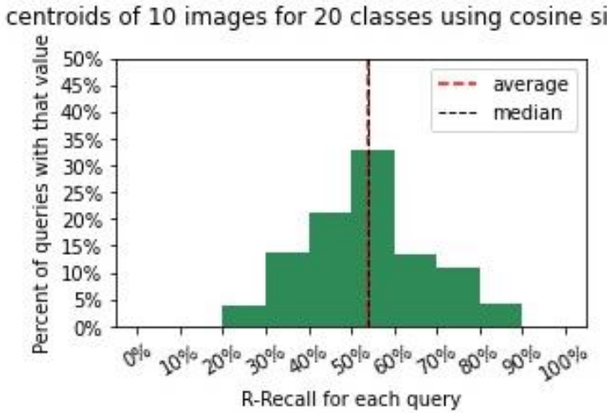
All class results

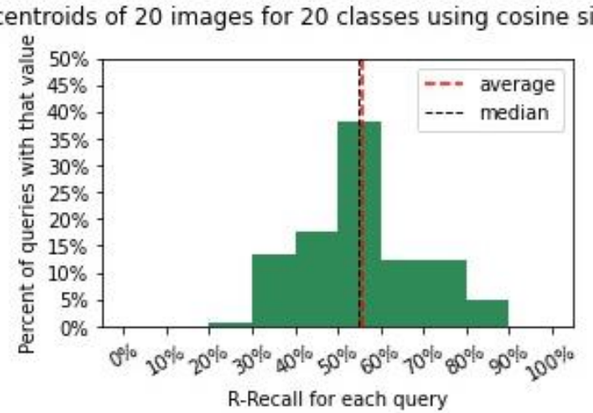
Aggregated across 20 classes

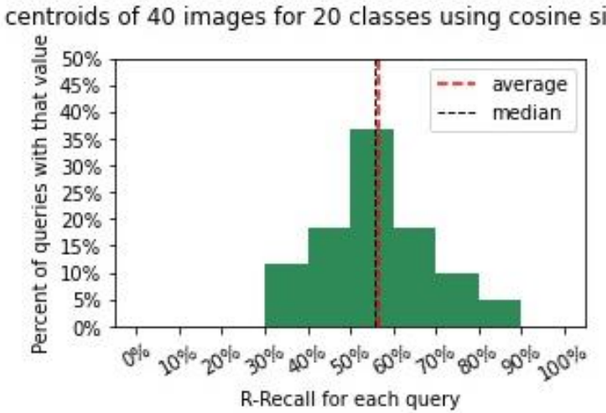












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Conclusions

- Centroid better than single image
- More images makes the centroid better
- More images give more consistency
- Only small improvements after 10-20 images

Impact

- Improvements without retraining
  - Inexpensive
- Improves the process of information gathering
  - Better
  - Still fast



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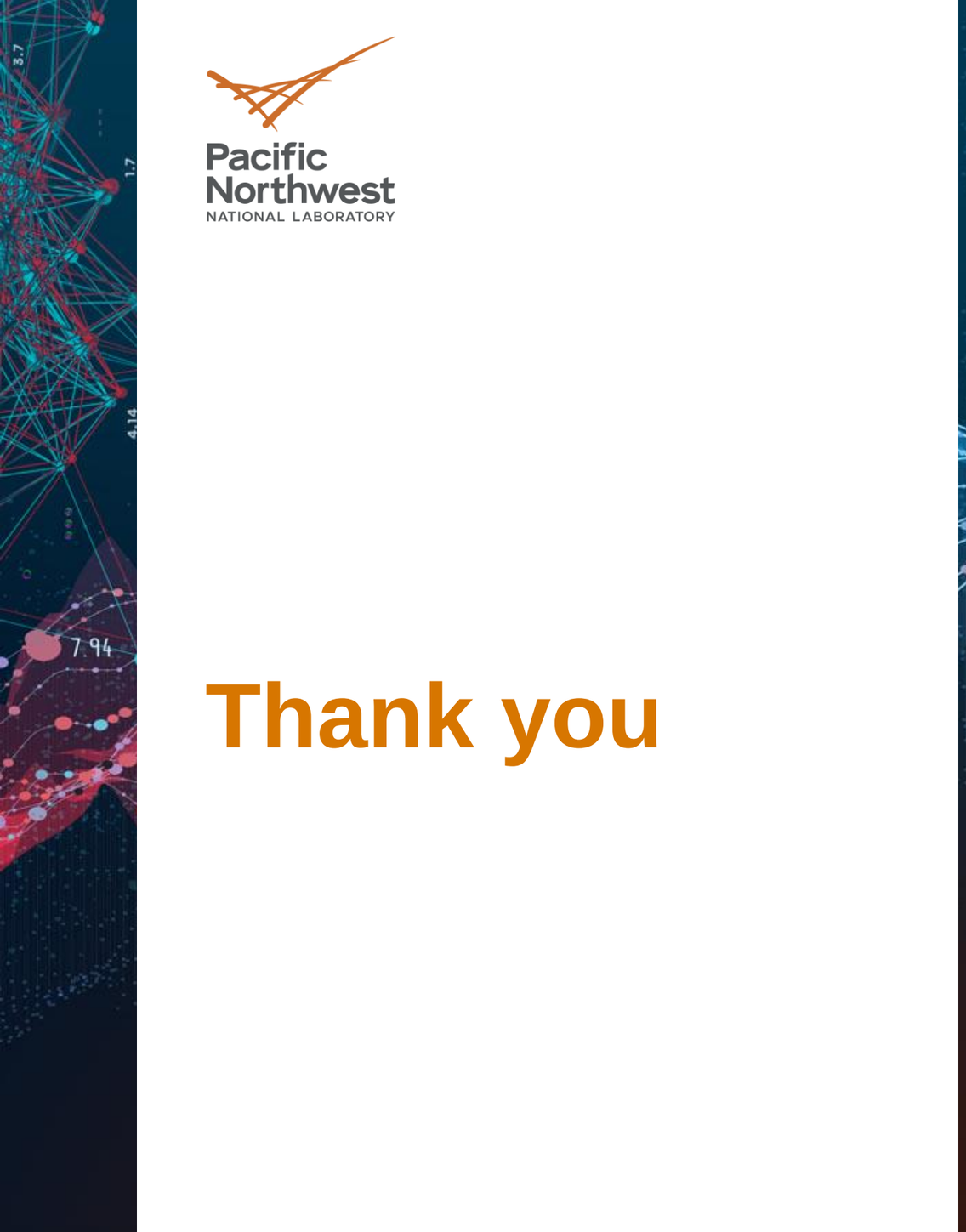
Acknowledgements

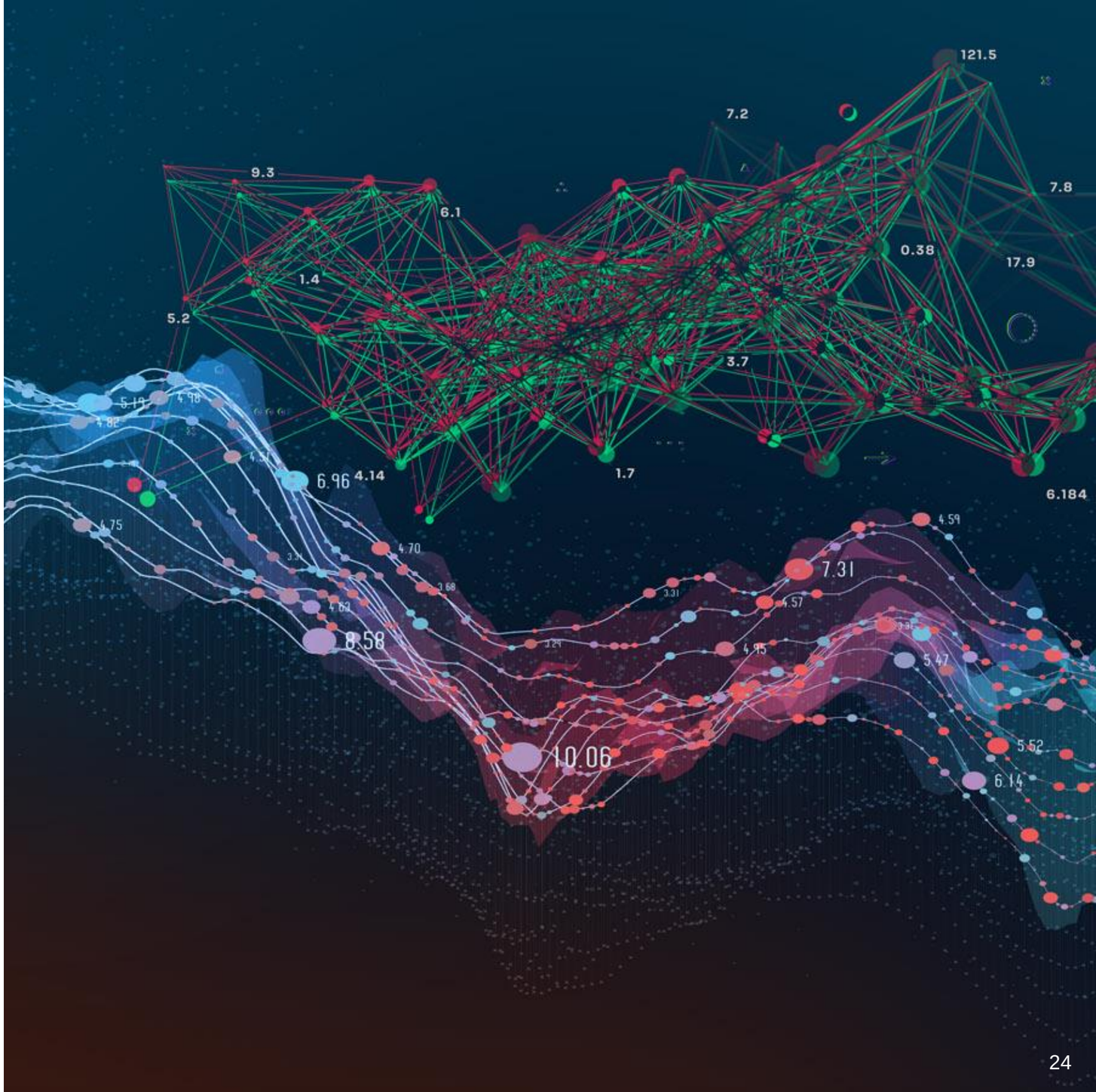
- Derek Tropf (Mentor)
- Melissa Swift (Co-mentor)
- Joshua Harrison
- Tegan Emerson
- Amanda White

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Thank you







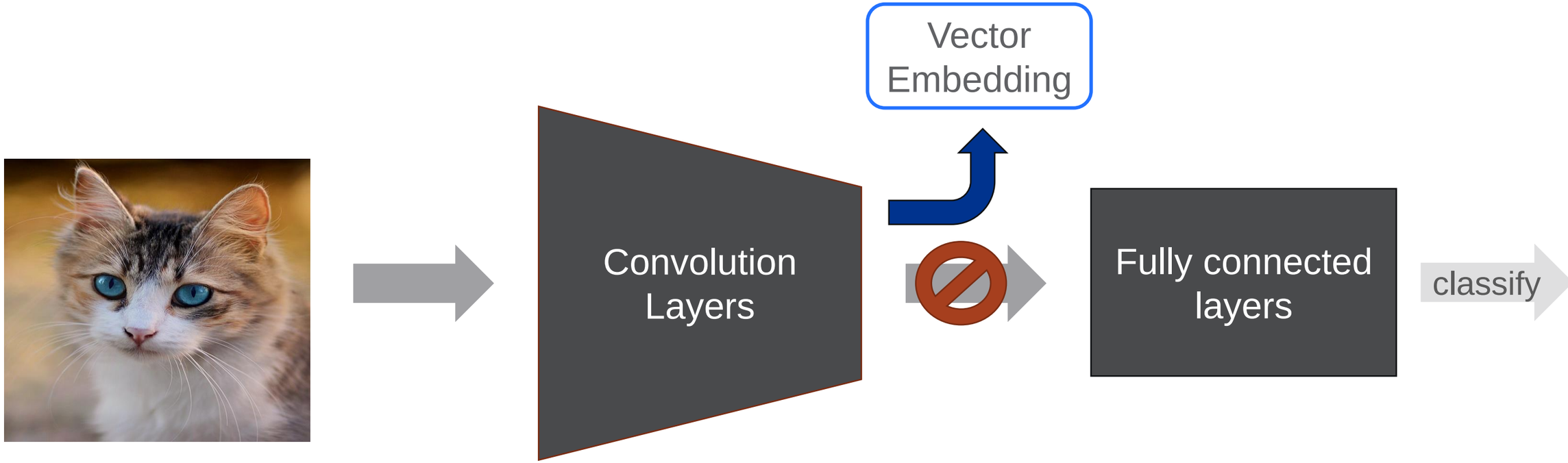
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Tools Used

- Opensearch
  - Docker
  - Amazon Web Services
- Xception model
  - Ran using AWS Lambda
- Python

How the vectors were produced

- Xception model
- Trained on ImageNet



- **Xception: Deep Learning with Depthwise Separable Convolutions,** F. Chollet, arXiv 2017, DOI: 10.48550/arXiv.1610.02357, https://doi.org/10.48550/arXiv.1610.02357
- J. Deng et al., "ImageNet: A large-scale hierarchical image database," 2009 IEEE Conference on Computer Vision and Pattern Recognition, 2009, pp. 248-255, doi: 10.1109/CVPR.2009.5206848.



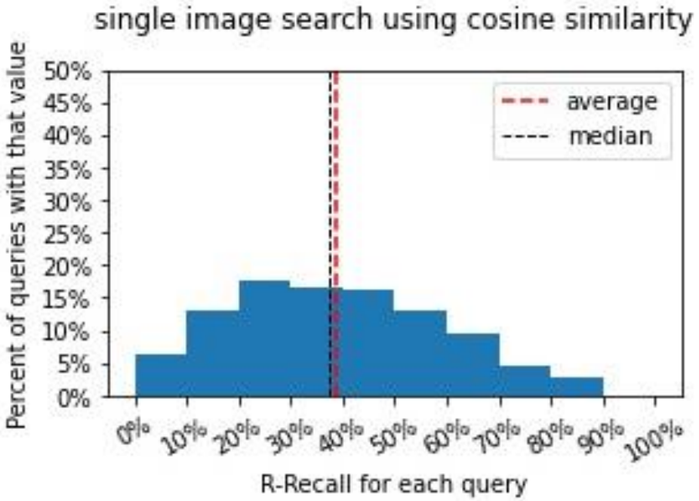
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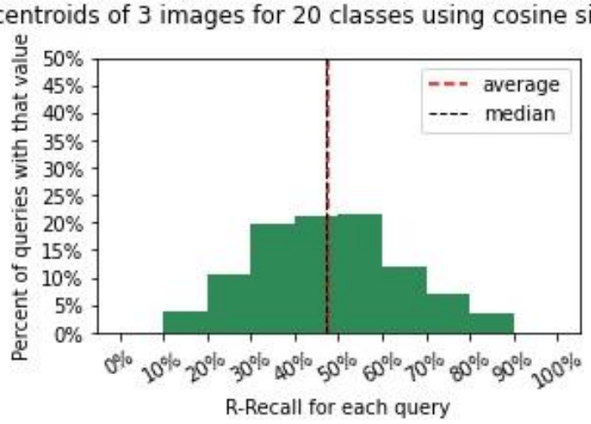
Further work

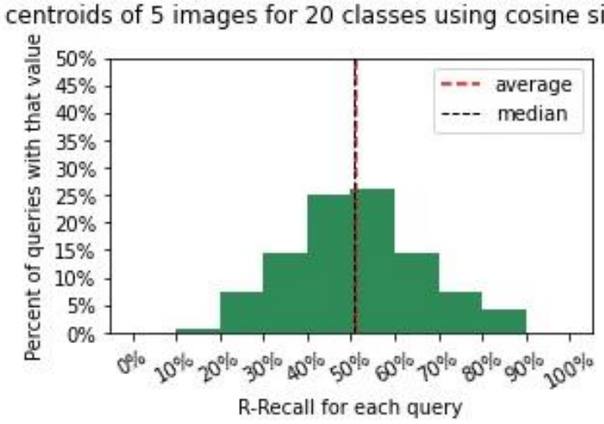
- Different datasets
- Different techniques
- Influence of image choice for centroid

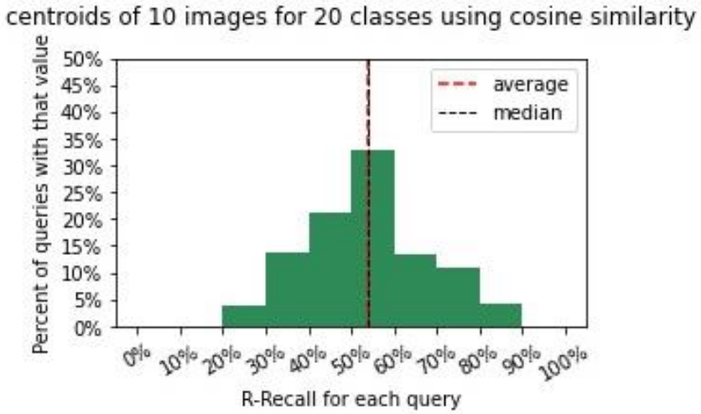
Searching with centroids of different k

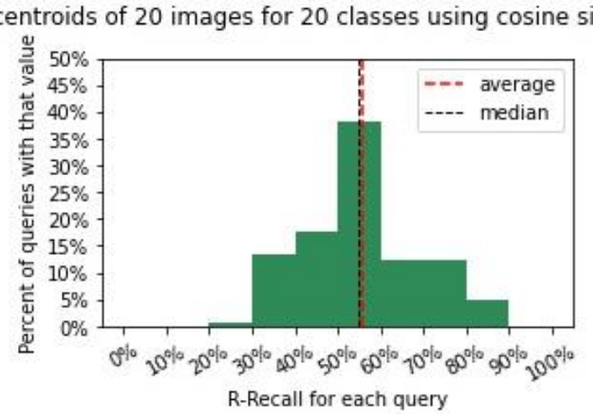
- Each is made from k images. Data is from cosine similarity

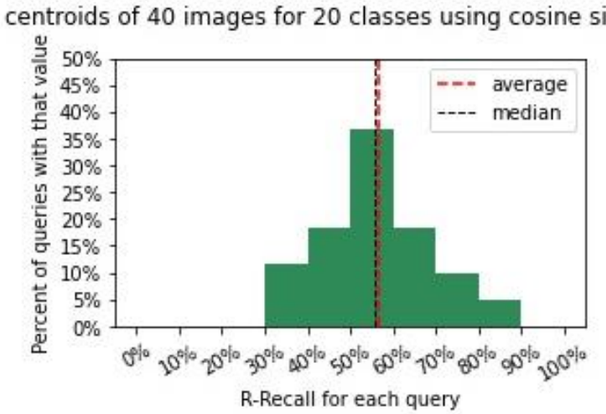






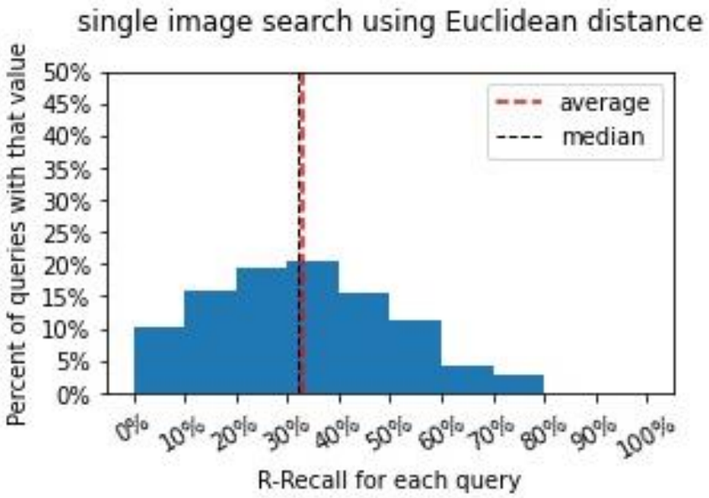


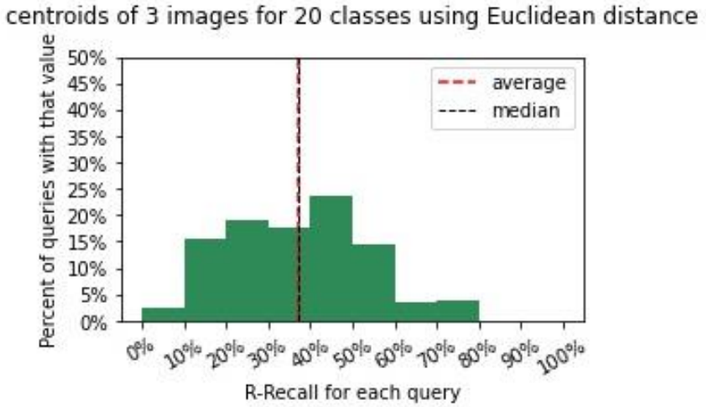


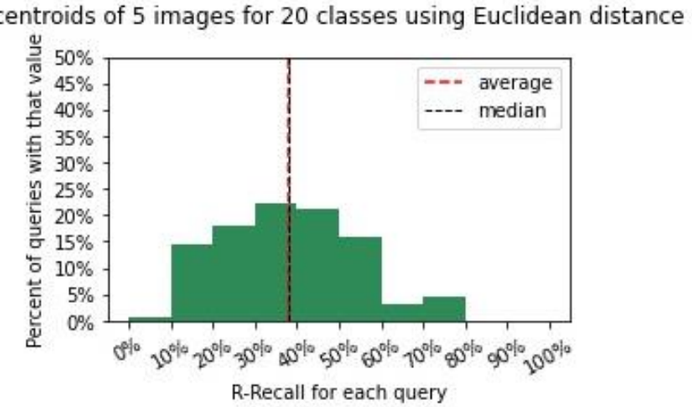


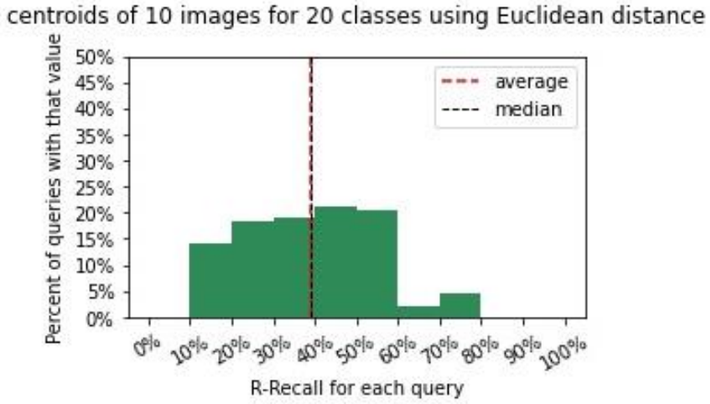
Searching with centroids of different k

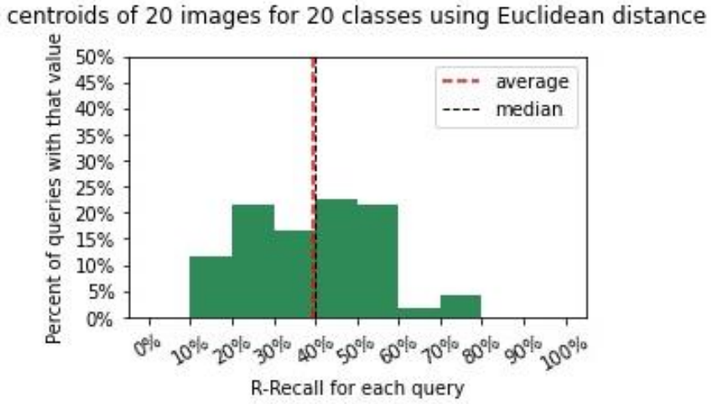
- Each is made from k images. Data is from Euclidean distance

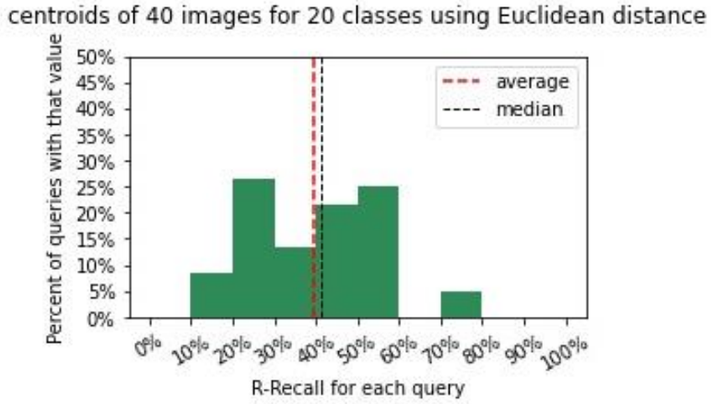






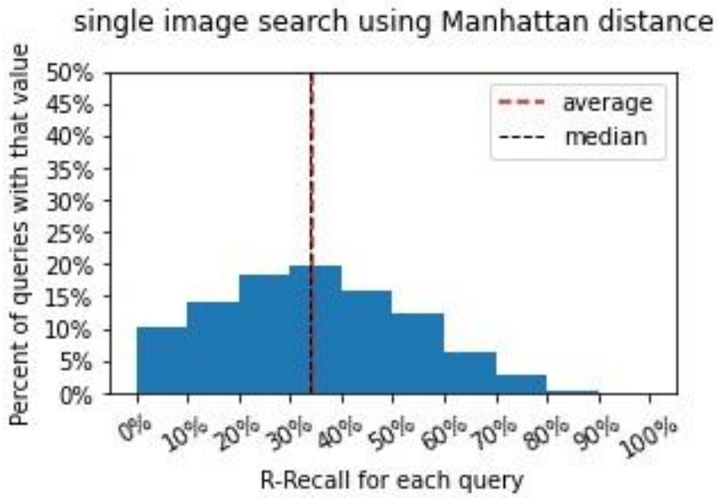


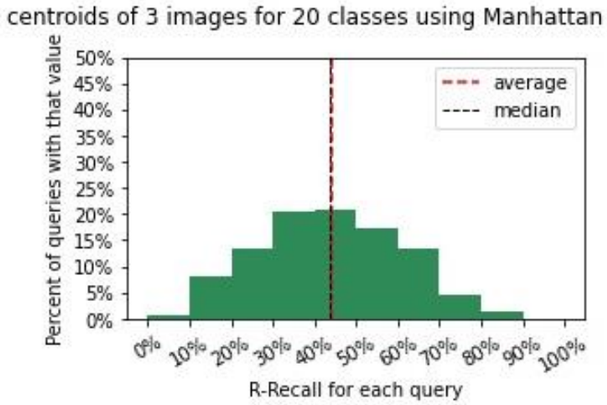


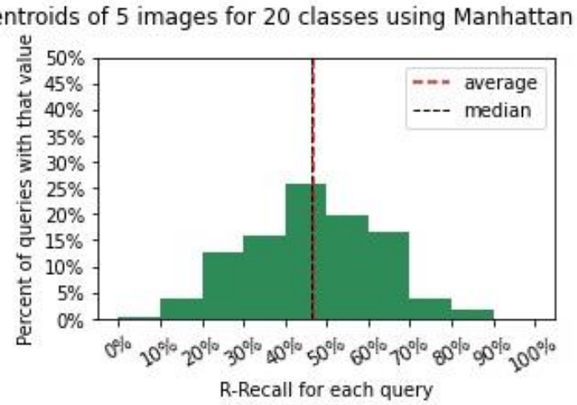


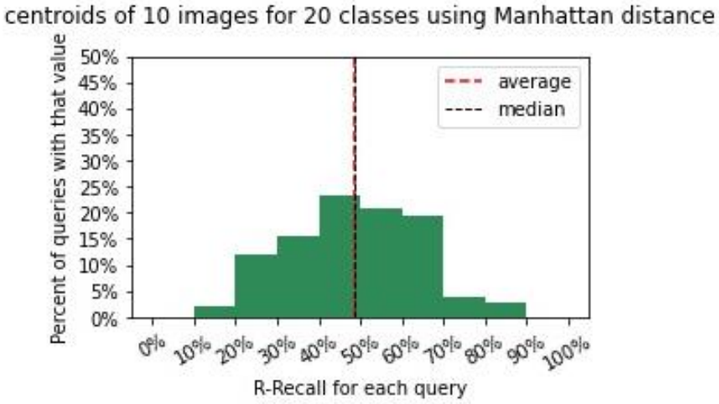
Searching with centroids of different k

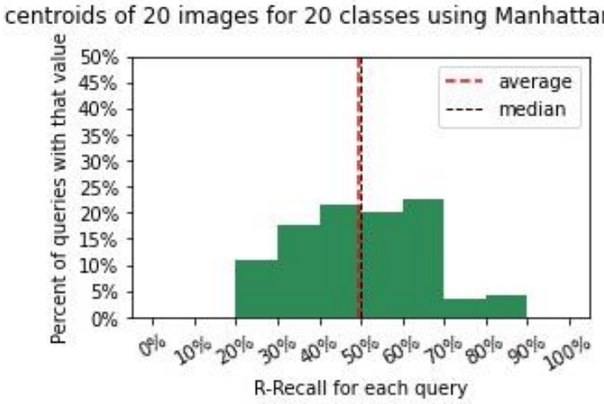
- Each is made from k images. Data is from Manhattan distance

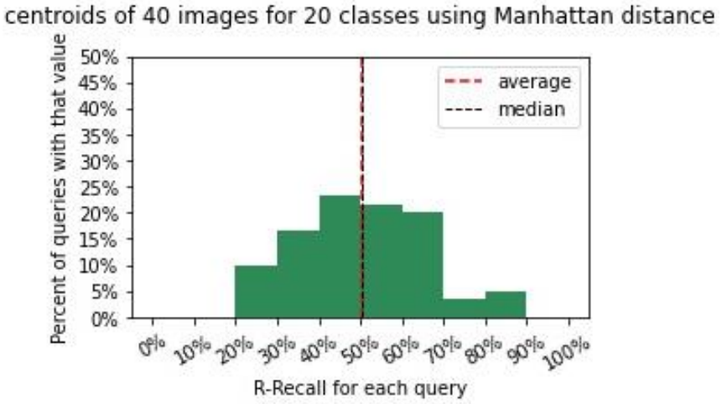






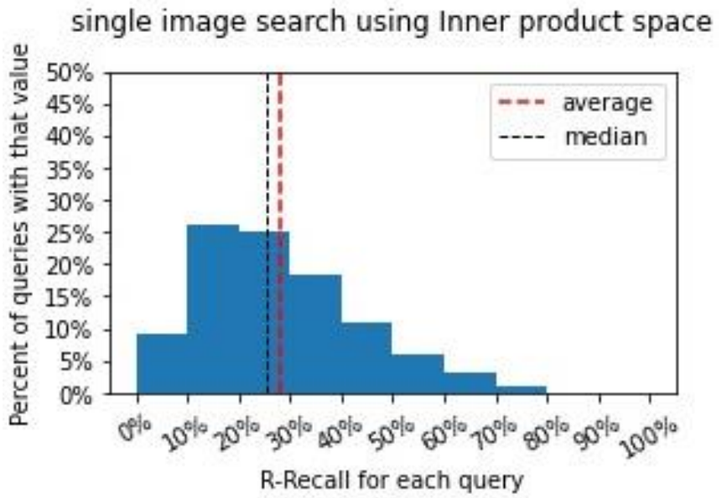


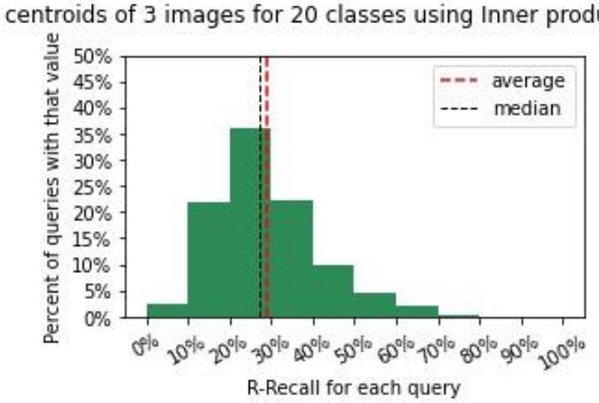


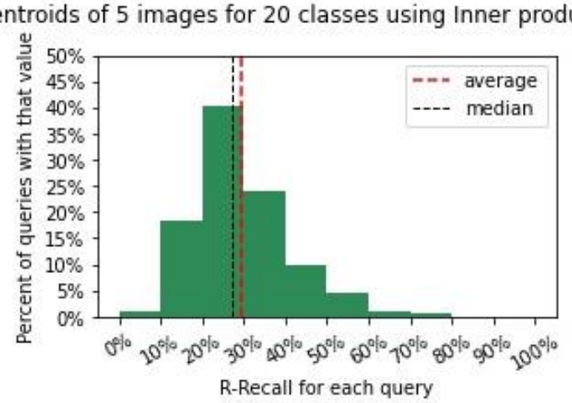


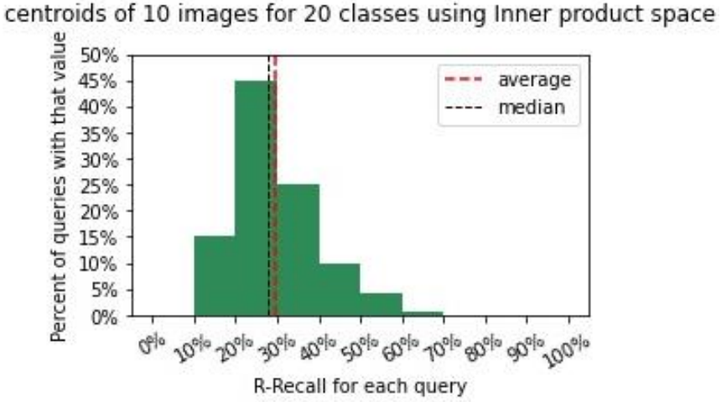
Searching with centroids of different k

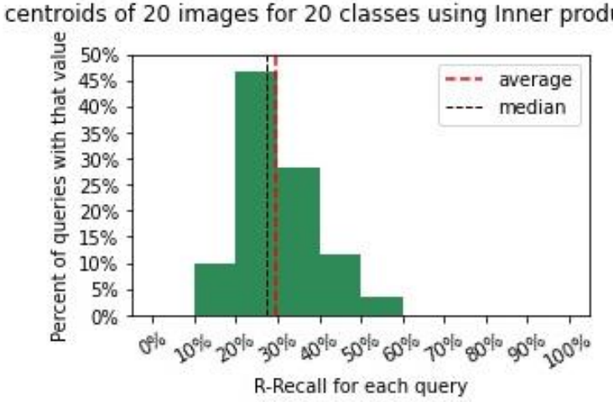
- Each is made from k images. Data is from inner product

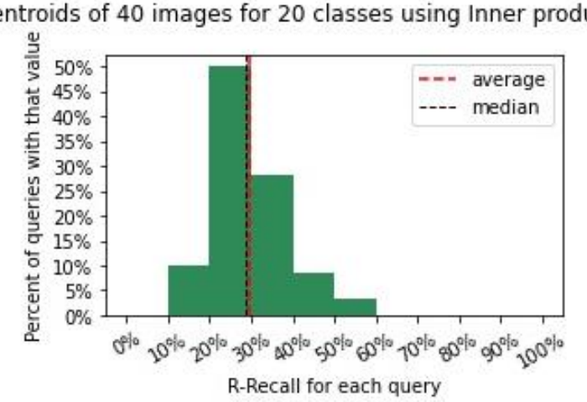






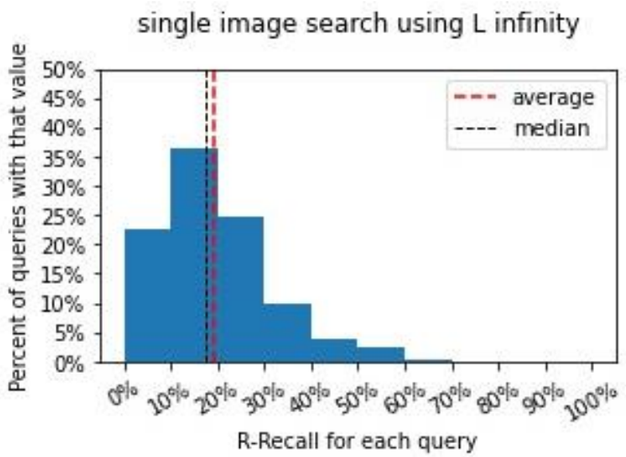


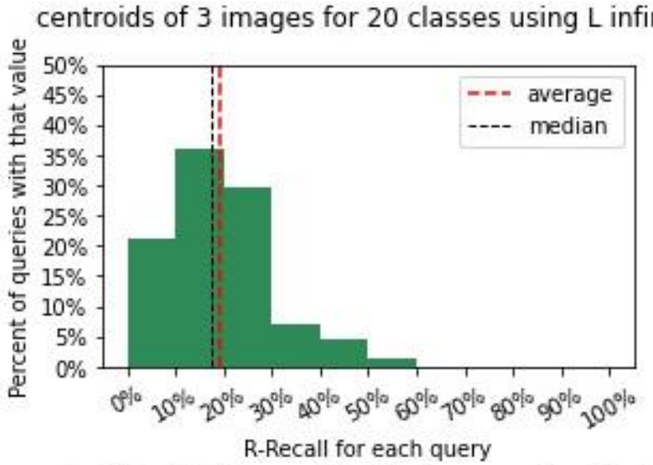


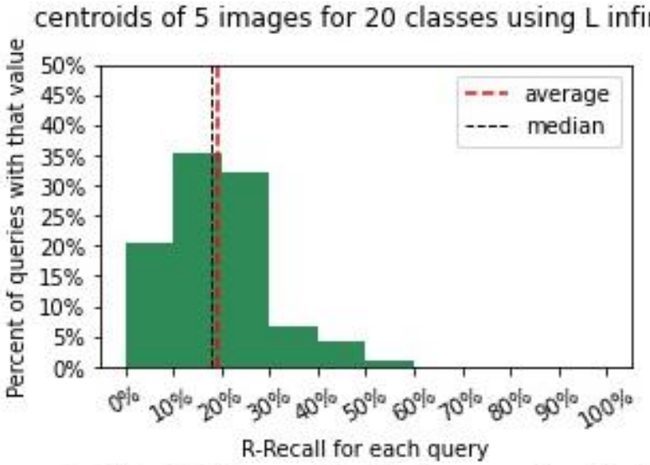


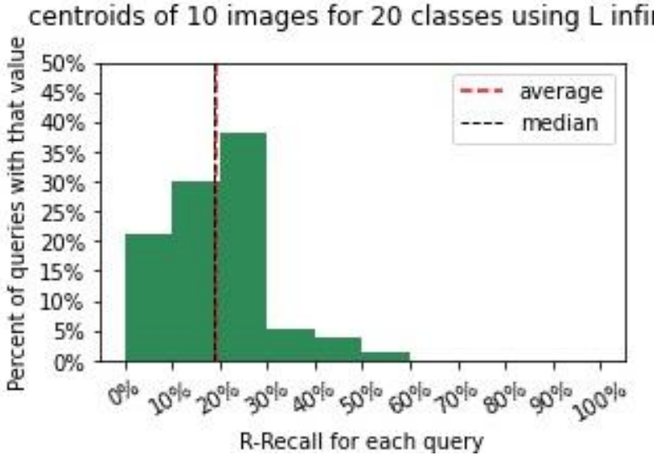
Searching with centroids of different k

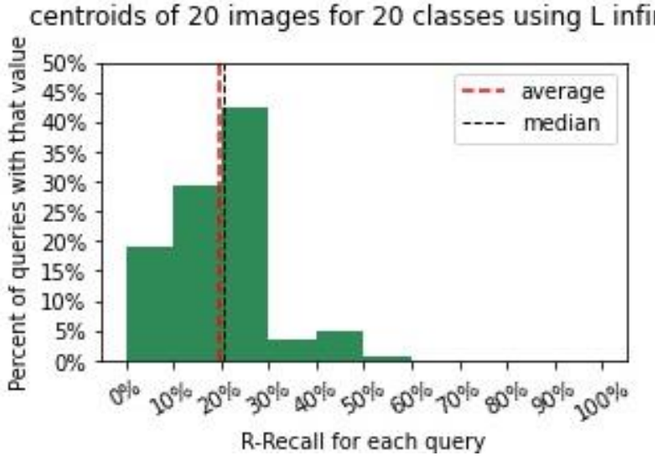
- Each is made from k images. Data is from L infinity

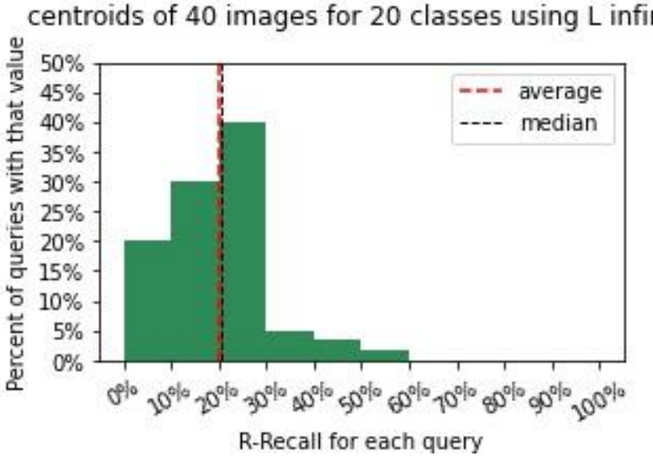












Dataset

Database Target images:

- 20 classes
  - Per class: 300 easy images[1],~30 hard images[2], ~12 trick images[2]

Search images:

- Same 20 classes
  - Per class: 124 easy images[1]

[1]https://www.image-net.org/ [2]https://github.com/hendrycks/natural-adv-examples

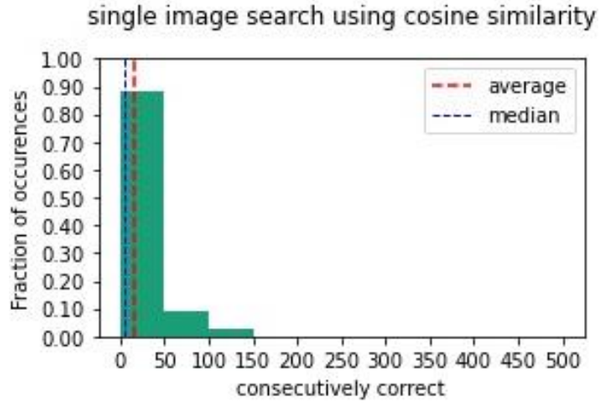
Test Method

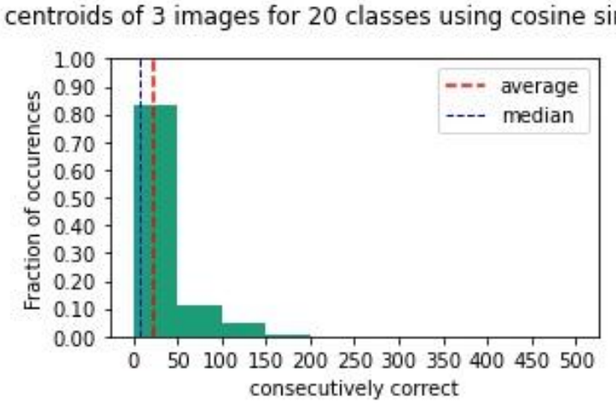
- For each class
- Search with every image
  - 124 searches per class
- Search with centroids
  - 124 ÷ number of images in centroid

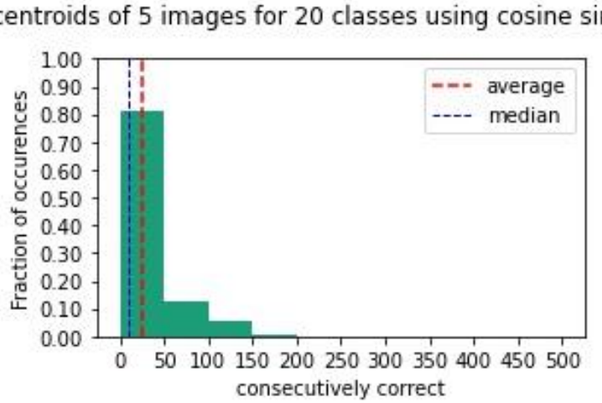
[1]https://www.image-net.org/ [2]https://github.com/hendrycks/natural-adv-examples

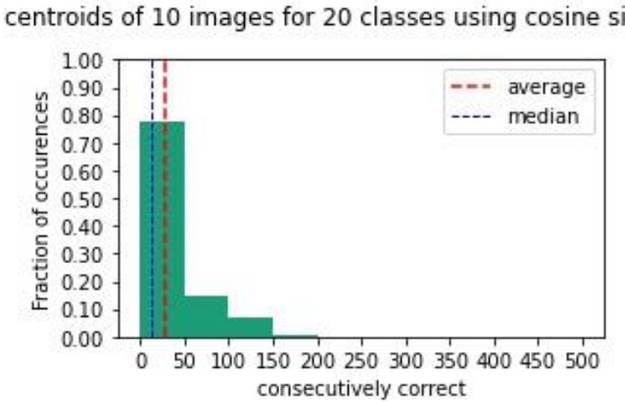
Results consecutively corrects

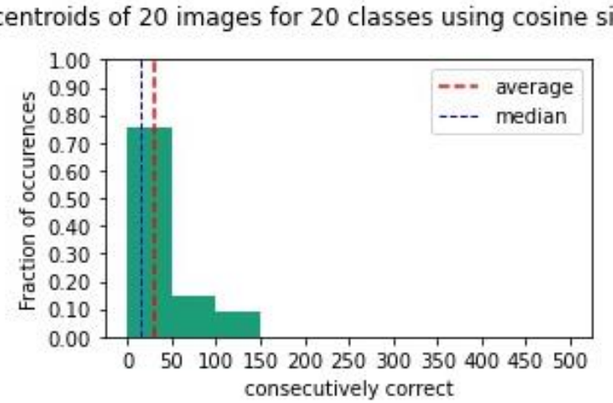
- Data is using cosine similarity. Aggregated across 20 classes

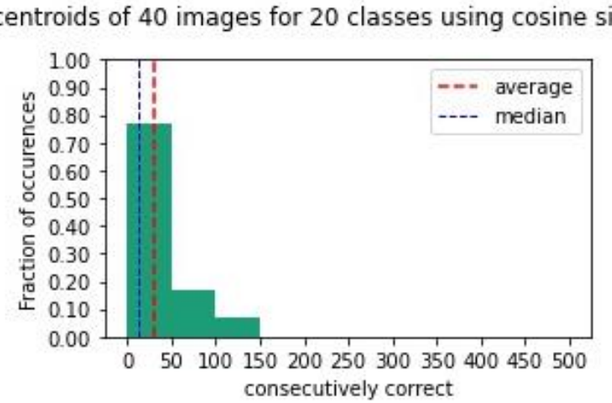












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Reverse image search using centroids of vector embeddings

December 15, 2022

Steven Harper
PNNL Intern



U.S. DEPARTMENT OF ENERGY BATTELLE

PNNL is operated by Battelle for the U.S. Department of Energy



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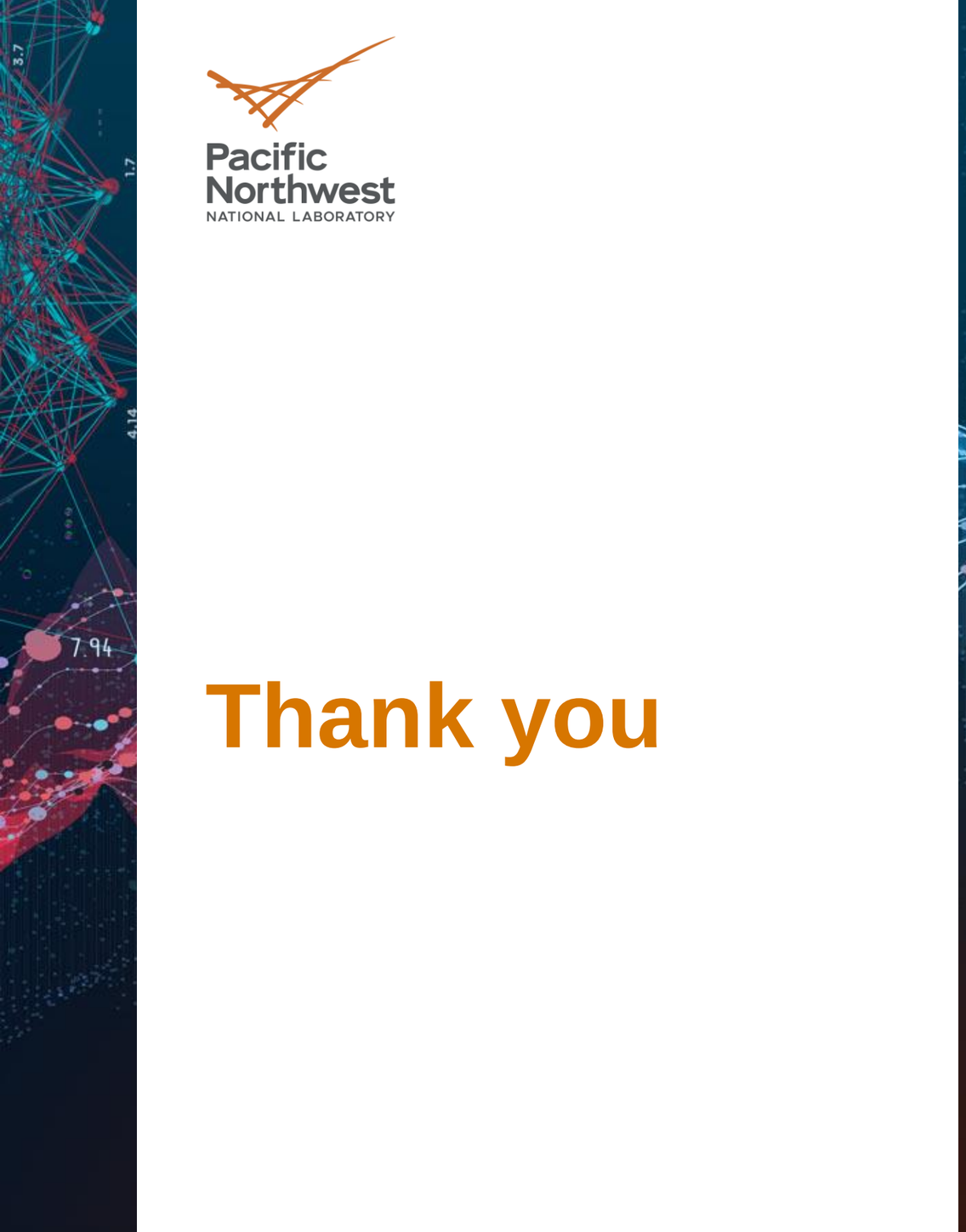

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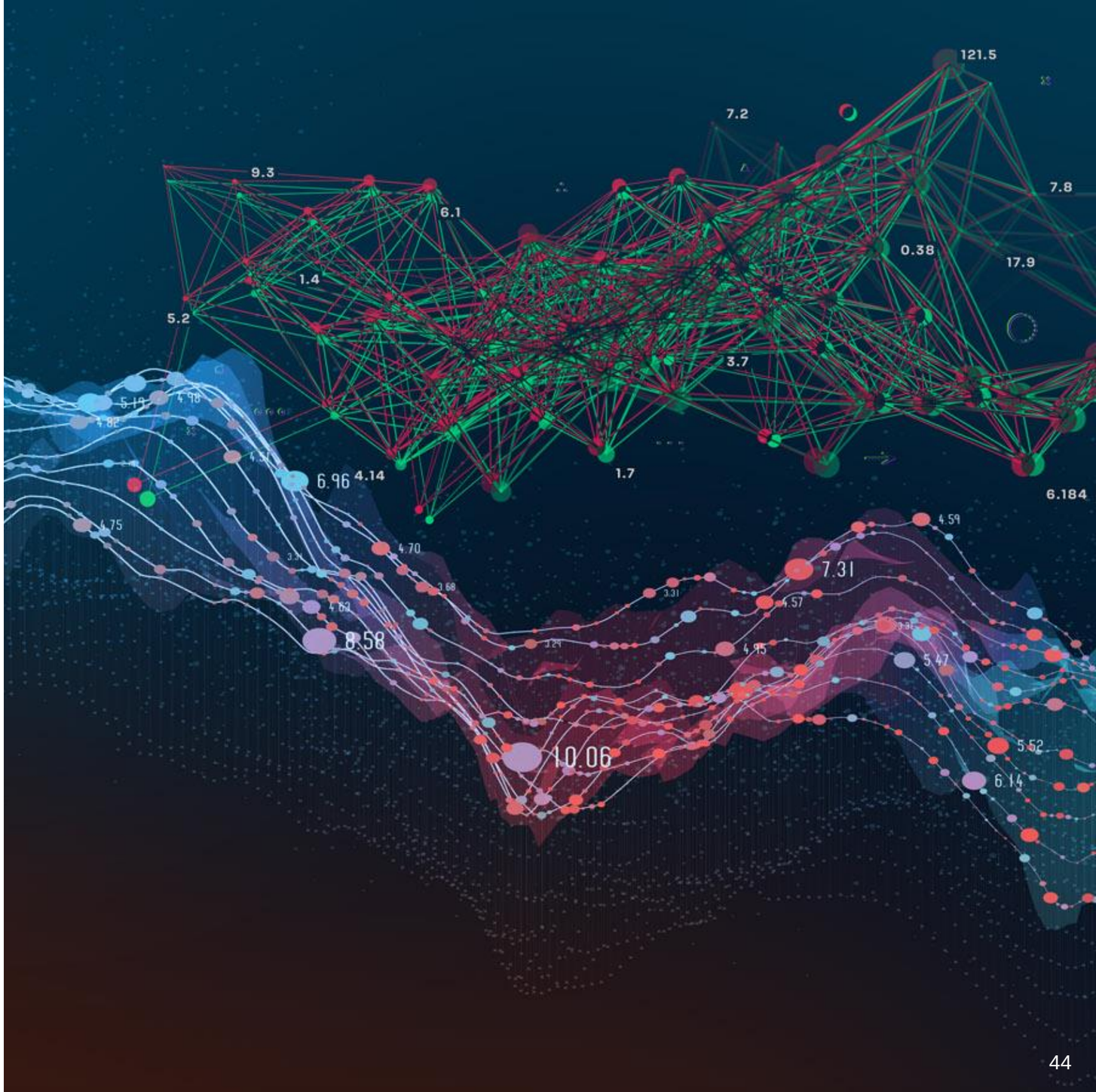


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Thank you





Test Method (improvement?)

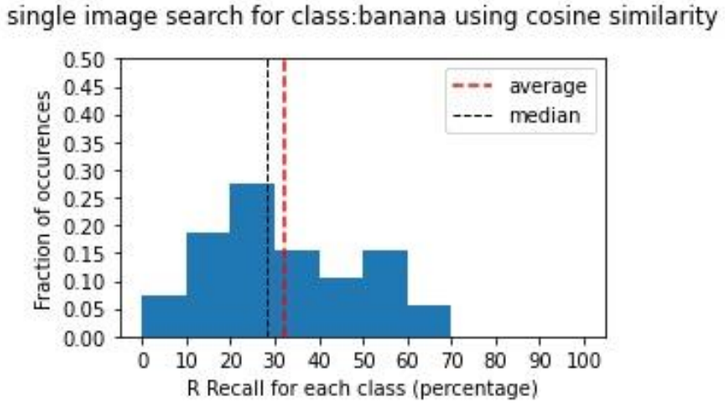
For each class

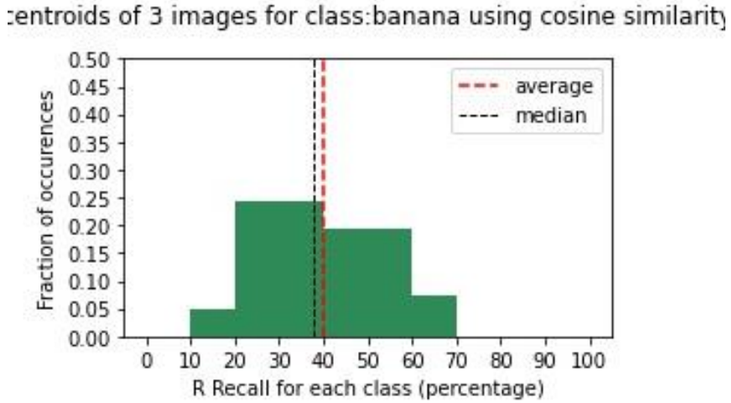
- Search with every image
  - 124 searches
  - Baseline

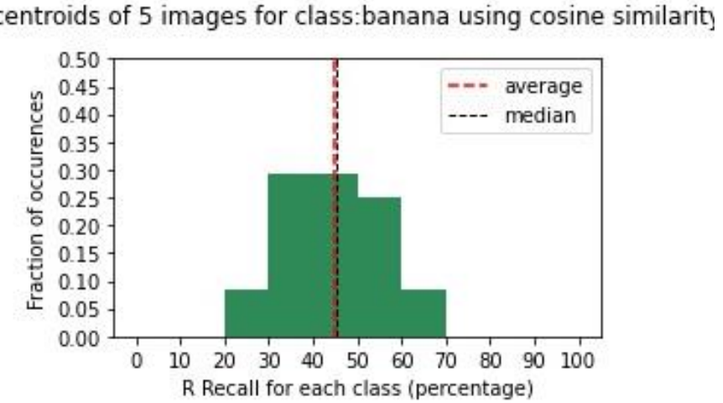
- Search with centroids
  - 124 ÷ number of images in centroid
  - Improvement?

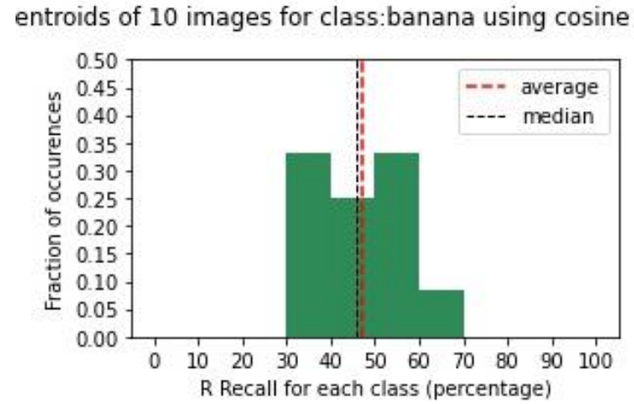
One class results

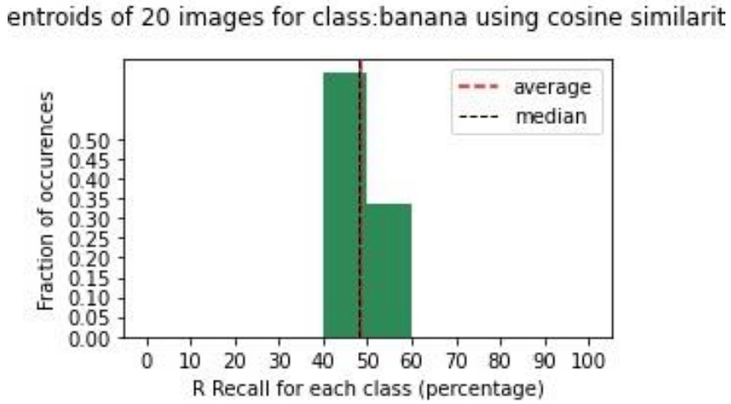
Banana class (not cats)

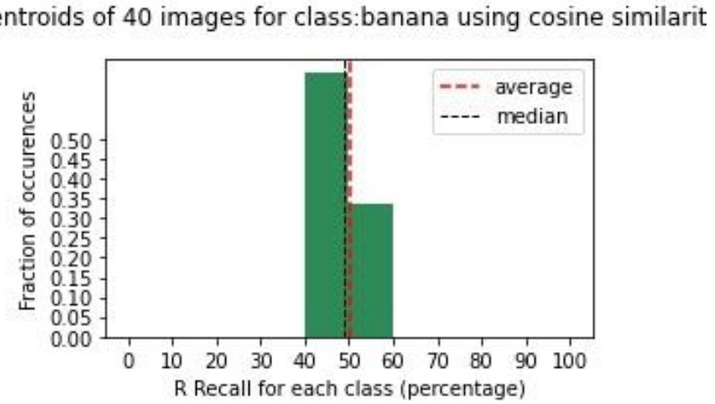














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Two different experiments

Compare:

- Single image searches
  - Baseline
- Centroid searches
  - Number of images per centroid
  - Improvement?



Model

Vector Embedding

[2,0,7,…,3,5,9]

Xception: Deep Learning with Depthwise Separable Convolutions, F. Chollet, arXiv 2017, DOI: 10.48550/arXiv.1610.02357, https://doi.org/10.48550/arXiv.1610.02357



Model

Vector Embedding

Xception: Deep Learning with Depthwise Separable Convolutions, F. Chollet, arXiv 2017, DOI: 10.48550/arXiv.1610.02357, https://doi.org/10.48550/arXiv.1610.02357

Dataset we used

- 20 different classes
- Each class was split in two
  - Target images (~340)
  - Search images (124)
- Each centroid had floor(124/N) trials

- J. Deng, W. Dong et al., "ImageNet: A large-scale hierarchical image database," 2009 IEEE Conference on Computer Vision and Pattern Recognition, 2009, pp. 248-255, doi: 10.1109/CVPR.2009.5206848.
- **Natural Adversarial Examples,** K. Z. Dan Hendrycks et al., arXiv 2021, DOI: arXiv:1907.07174, https://arxiv.org/abs/1907.07174

How to measure effectiveness? Recall

- Recall = found cats ÷ all database cats

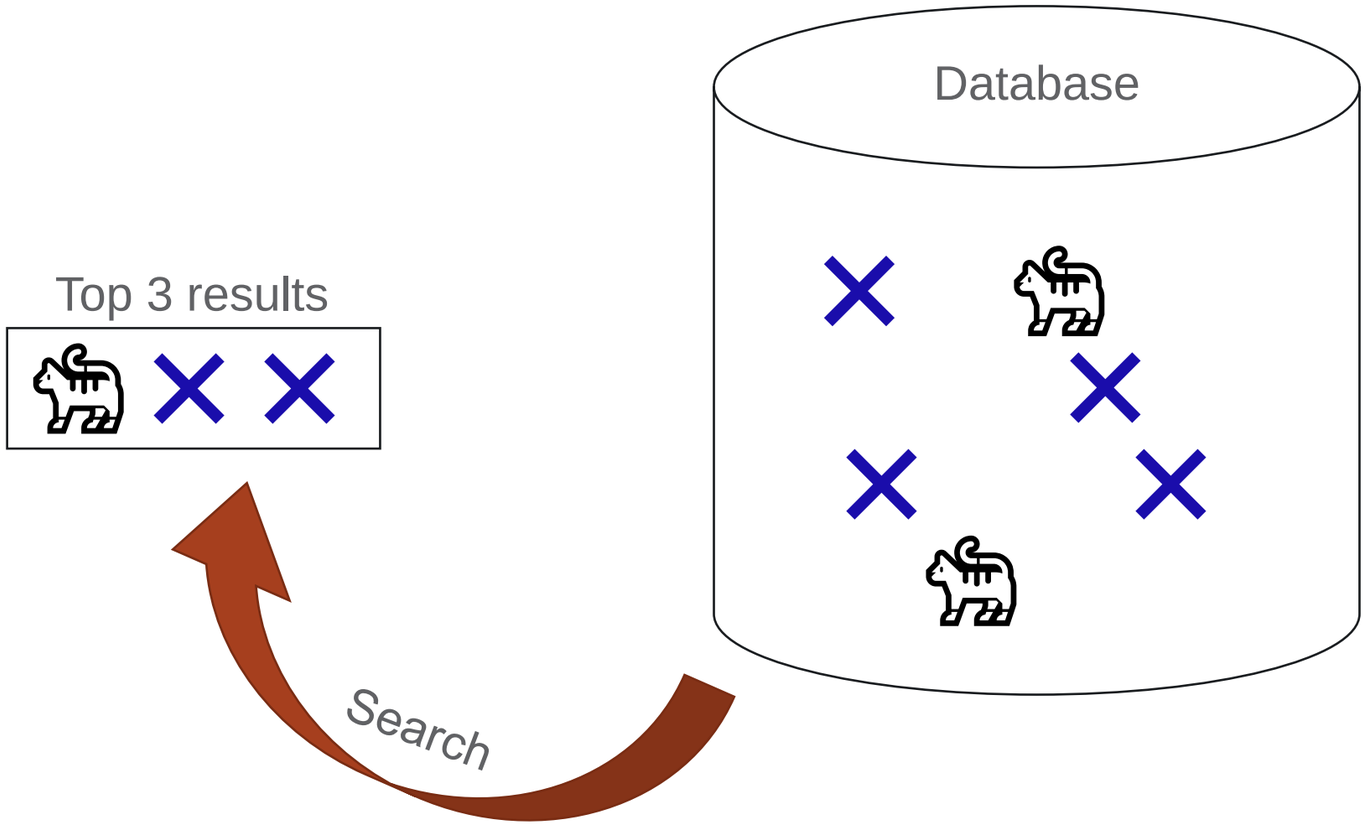
- 100% -> found every cat
- 0% -> found no cats

- Usable when target images are known

Evaluation measures (information retrieval), Published by: Wikipedia, Last Updated: 29 November 2022, https://en.wikipedia.org/wiki/Evaluation_measures_(information_retrieval)#Recall

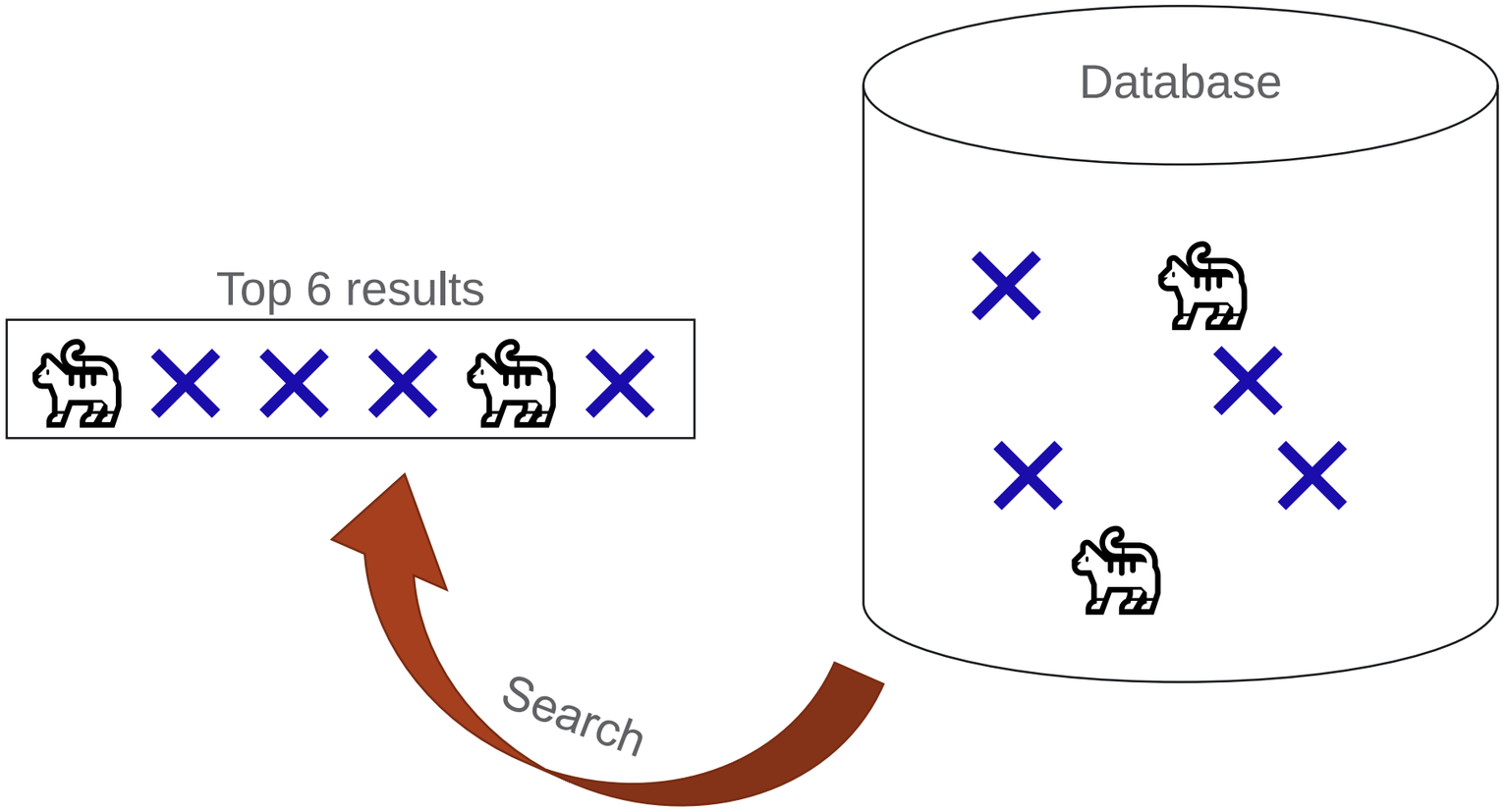
Recall Example

- Recall = 1 found ÷ 2 possible = 50%



Problems with recall

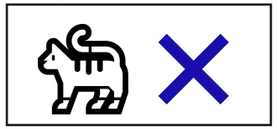
- Return all images -> always 100% recall



R-Recall (modification of Recall)

- Number of results = number of database cats
- 2 cats in the database, so only check top 2 results

Top 2 results



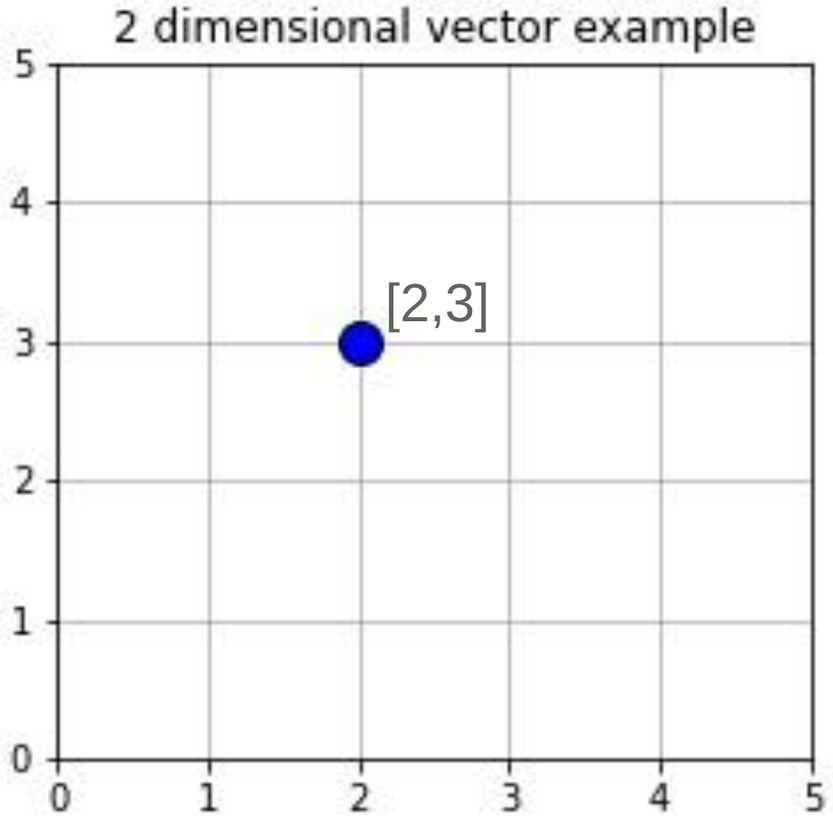
54

Evaluation measures (information retrieval), Published by: Wikipedia, Last Updated: 29 November 2022, https://en.wikipedia.org/wiki/Evaluation_measures_(information_retrieval)#Recall

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What is a vector?

- Just a list of numbers
- Each number is a dimension



- Another example is [1,3,8,7] (4 dimensions)
- Our vectors have 2048 dimensions!

What is a vector embedding?

- Vector with embedded meaning
- Describes the image contents
- Model’s interpretation
- Less data than the image



[2,0,3,6,0,8,2,5, … ,7,8,2,84,0]